Labeling Chaos to Learning Harmony: Federated Learning with Noisy Labels

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01 Motivation

The problem

Federated learning (FL) must learn from unreliable client labels despite uneven noise, limited supervision and unequal device resources.

Contributions

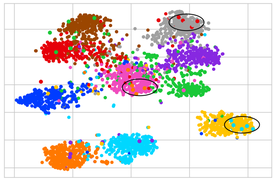
- **Client-specific noise estimates** identify uneven supervision quality across devices.
- **Three alternative interventions** match correction, teacher guidance or aggregation to device resources.
- **Broad evaluation** tests synthetic vision/audio noise and human-annotated image noise.

Correct labels before training

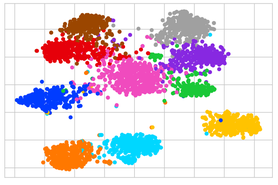
Nearest neighbor-based correction (*NNC*) uses local embedding neighbors to correct class labels before training. Input examples and embeddings stay fixed.

Before and after label correction

Before NNC



After NNC



CIFAR-10; 40% random noise. Fixed contrastive language-image pre-training (CLIP) features.

Scope

Pretrained encoders add cost; compute is unmatched and three-run results omit dispersion. *NNC* can fall below FedAvg under class flipping.

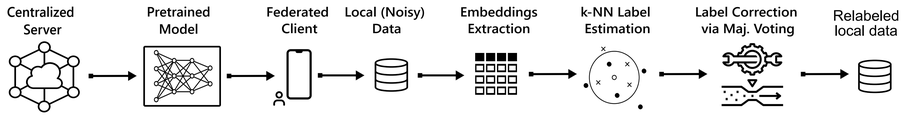
Notation

adaptive knowledge distillation (*AKD*) · noise-aware federated averaging (*NA-FedAvg*) · knowledge distillation (KD) · k-nearest neighbors (kNN) · confident learning (CL) · percentage points (pp) · cross-entropy (CE)

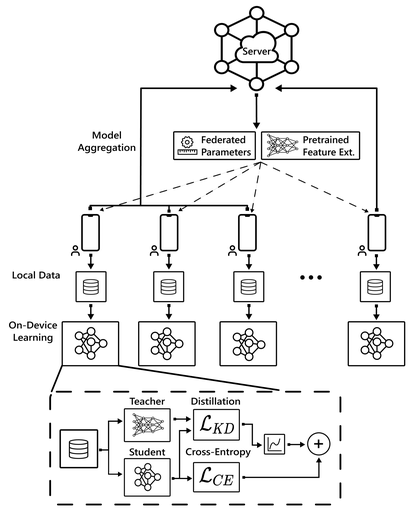
02 FedLN

Three alternative interventions address client label noise

INITIALIZATION · NNC

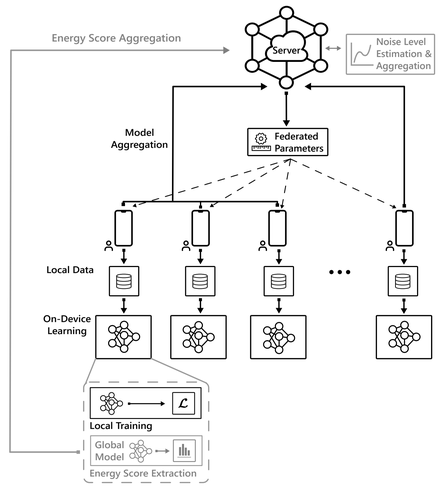


LOCAL TRAINING · AKD



AGGREGATION · NA-FedAvg

Estimate noise once after warm-up



Technical core

$$\mathcal{L}_{\text{AKD}} = \mathcal{L}_{\text{CE}} + \beta u_{\varepsilon}(n_l^m)\mathcal{L}_{\text{KD}}$$

For client m, estimated noise n_l^m gates teacher guidance through u_ε. β controls its strength alongside label-based CE.

$$\gamma_m = (1 - n_l^m)\frac{N_m}{N}$$

NA-FedAvg multiplies client m's data share N_m/N by its estimated clean-label fraction $1\text{-}n_l^m$.

Paper

Code

03 Evaluation

+17.11 pp

NNC vs. FedAvg

CIFAR-10; 70% random noise

+10.14 pp

AKD vs. FedAvg

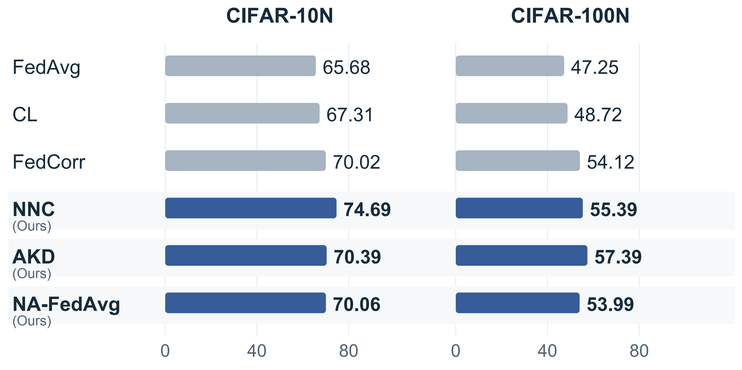
CIFAR-100N; human label noise

Accuracy depends on the noise pattern

Method	Random noise (%)	Class flipping (%)
FedAvg	57.65	63.54
FedCorr	67.27	65.23
NNC (Ours)	**74.76**	**54.59**
AKD · embeddings (Ours)	**68.43**	**68.39**
NA-FedAvg (Ours)	**65.34**	**67.92**

CIFAR-10/ResNet-20; 30 clients, 200 rounds, three-run mean accuracy. 70% label noise on 80% of clients; random versus class-flipping noise.

Accuracy with human annotation noise



Three-run mean test accuracy (%); dispersion unreported. 30 clients, 80% participation; 200/500 rounds for CIFAR-10N/100N. *AKD* uses embeddings.

Takeaway

Match the intervention to the noise: *NNC* gains 17.11 pp under random noise but falls below FedAvg under class flipping. With human annotations, **NNC leads on CIFAR-10N and AKD on CIFAR-100N** among these methods.